

GEFFEN COOPER

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EDUCATION

The University of Texas at Austin

Aug. 2022 - present

Ph.D. in Electrical and Computer Engineering (GPA: 3.9/4.0)

- **Advisor:** Radu Marculescu (Systems Level Design Group)

University of California, Santa Barbara

Sept. 2018 - June 2022

B.S. in Computer Engineering (GPA: 3.97/4.0)

- Regents Scholar

RESEARCH INTERESTS

Energy Harvesting, Batteryless IoT, Efficient Deep Learning, TinyML, Human Activity Recognition

RESEARCH PUBLICATIONS

Geffen Cooper, N. Tsao, F. Fotiadis, U. Topcu, and R. Marculescu, “BHAR: A System-level Optimization Framework for Multisensor **B**atteryless **H**uman **A**ctivity **R**ecognition,” in ***UNDER REVIEW*** at 2026 IEEE/ACM International Conference on Codedesign of Embedded Systems (CODES).

Geffen Cooper, Radu Marculescu, “Batteryless Gesture Recognition via Learned Sampling,” IEEE International Conference on Body Sensor Networks (BSN 2025).

Geffen Cooper, Radu Marculescu, “Packet Pruning: Finding Better Energy Spending Policies for Batteryless Human Activity Recognition,” IEEE International Conference on Body Sensor Networks (BSN 2024).

Geffen Cooper, Radu Marculescu, “Beyond Thresholds: A General Approach to Sensor Selection for Practical Deep Learning-based HAR,” IEEE/ACM International Conference on Internet-of-Things Design and Implementation (IoTDI 2024).

Geffen Cooper, Tianda Huang, Radu Marculescu, “Demo Abstract: A Prototype for Machine Learning with Batteryless Sensors,” IEEE/ACM International Conference on Internet-of-Things Design and Implementation (IoTDI 2024).

WORK EXPERIENCE

Wiliot

June 2025 - May 2025

Data Scientist

Remote

- Feature engineering and training for an XGBoost-based classifier to detect if an asset was discarded using RSSI and location data from Wiliot’s batteryless BLE pixels
- Built a performance monitoring tool using retrospective data to evaluate an online model without access to ground truth

Analog Devices

June 2022 - August 2022

Embedded AI Applications Intern, Embedded AI Team

Dallas, TX

- Implemented a few-shot domain adaptation technique to fine-tune CNN models for the MAX78000 microcontroller with onboard CNN accelerator
- Adapted pretrained models using 5-10 images per class from the onboard camera to improve classification accuracy on camera images by 20-40%

USC SURE, Institute for Creative Technologies

June 2021 - September 2021

Undergraduate Research

University of Southern California (Remote)

- Worked under Professor Mohammad Soleymani in the Intelligent Human Perception Lab
- Developed a multimodal deep learning model to assess psychomotor retardation from audio-visual recordings of clinical interviews

Brain Corp

June 2020 - September 2020

*Embedded Software Intern, Firmware Team**San Diego, CA (Remote)*

- Created a robust Linux based C/C++ application using socketCAN to update the firmware of the vehicle controller used in Brain Corp's fleet of autonomous floor scrubbing robots

FLIR Systems

April 2019 - August 2019

*Software Intern, Security Team**Goleta, CA*

- Wrote bash scripts and python GUI for running diagnostics, debugging, and configuring thermal cameras

COURSEWORK

Convex Optimization, ML for Real World Networks, Digital Video Processing, Human Signals, Probability and Stochastic Processes, Applied ML, Generative AI, Embedded IoT, Learning Based Optimal Control

SKILLS

Python, C, Bash, PyTorch, TensorFlow, TensorFlow Lite, Git

ADDITIONAL PROJECTS**Few-Shot Gesture Recognition**

Human Signals Course 2023

- Trained an LSTM with attention using sequences of 3D hand landmarks and used embeddings at inference time to classify new gestures from 3-5 examples

Graph Neural Network Compression

Machine Learning for Networks Course 2022

- Used graph partitioning and knowledge distillation to reduce memory during inference for node classification

TinyML Projects with Max78000

Collaboration with Analog Devices 2021-2022

- Deployed image classification and detection models with MAX78000 AI microcontroller with CNN accelerator
- **G. Cooper**, V. Benenati, B. Long, K. Copeland, T. Ekaireb, S. Kumar, B.S. Manjunath, Y. Isukapalli, "Autonomous System for Sorting Objects at The Edge," Proceedings, International Telemetering Conference (ITC), Glendale, Arizona, Oct 24-27, 2022.
- **G. Cooper**, B.S. Manjunath, and Y. Isukapalli, "Edge Machine Learning for Face Detection," Proceedings, International Telemetering Conference (ITC), Las Vegas, Nevada, Oct 25-28, 2021.